Chapter 10: Factorial ANOVA

Statistics for Environmental Science

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In science, outcomes rarely depend on just one thing. Climate, soil type, and water availability all act together to influence ecosystems in ways a single-factor design can't capture. *Factorial designs* let us study several factors, and how they interact with each other, at once.

Chapter 8's one-way ANOVA tested a single factor, the ANOVA name for an independent variable (IV). This chapter uses both words. This chapter covers how to compute a factorial ANOVA, including a full by-hand walkthrough of the sums-of-squares arithmetic, and how to interpret the result. With more than one factor, interpretation is harder, because an *interaction* changes what the main effects in the results table mean.

Factorial basics

Imagine investigating the growth rate of a plant species, with IVs like soil pH or water availability. Each added IV multiplies the number of conditions you can observe, and with it, the questions you can ask about how those factors work together. Table 1 shows three designs built around the same plant-growth question.

Table 1: The test depends on how many IVs there are and how many levels each has.

Design	Example question	Test
1 IV, 2 levels (soil pH: acidic, neutral)	How does soil pH affect plant growth?	t-test
1 IV, 3 levels (water: low, medium, high)	How does water availability affect plant growth?	One-way ANOVA
2 IVs, 2 levels each (soil pH acidic or neutral × water low or high)	How do soil pH and water availability together affect plant growth?	Factorial ANOVA

Now we have two IVs, soil pH and water availability, each with two levels. A **factorial** design has two or more IVs, and each of its conditions is a combination of one level from each IV.

A **fully factorial** design includes every combination of the levels of its IVs. A design with this property is also called fully crossed. A factorial design that leaves some combinations of levels unmeasured is only partly crossed. Every design in this chapter is fully factorial, so the rest of the chapter uses factorial to mean fully factorial.

Factorial notation

Each combination of soil pH and water availability is one condition. With 2 IVs, each with 2 levels, there are $2 \times 2 = 4$ conditions in which to measure growth rate.

The name of a factorial design gives one number per IV, and each number is that IV's count of levels. Multiplying the numbers gives the number of conditions. Here are a few examples:

- 2x2: Two IVs, each with two levels (yields four unique conditions).
- 2x3: Two IVs, with the first having two levels and the second three (yields six unique conditions).
- 3x2: Similar to the 2x3, but the first IV has three levels, and the second has two, also giving us six unique conditions.
- 4x4: Two IVs, each with four levels, resulting in sixteen conditions.

Pollution, flow, and stream life

The rest of this chapter uses one example throughout.

The question. Streams are affected by pollution, but also by how fast the water moves. Fast-moving water flushes contaminants downstream and re-oxygenates the channel. Slow water lets them sit. So pollution might do more damage in a sluggish stream than a fast one. To measure that damage, we need to operationalize stream health.

The dependent variable is macroinvertebrate species count, the number of distinct kinds of aquatic insects, snails, worms, and crustaceans that live in a stream. This is a standard measure of stream health. These animals are in the middle of the food web, many are pollution-sensitive, and they can't flee a degraded stream the way fish can. More species present means a healthier stream.

The design. We run an experiment in artificial stream channels, each assigned one pollution condition and one flow rate. The species counts in this chapter are invented, chosen so the arithmetic stays easy to follow. Figure 1 summarizes it.

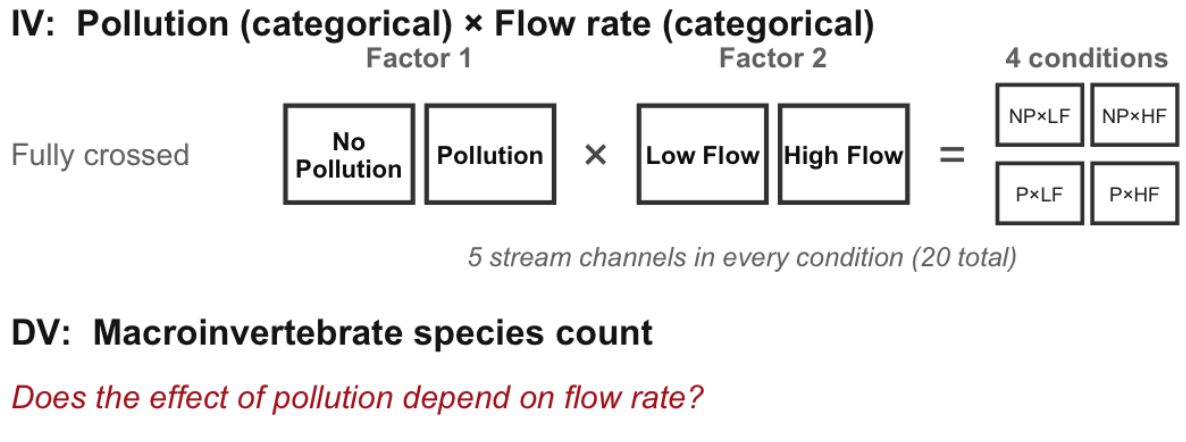


Figure 1: Study design for the stream experiment. Two pollution levels crossed with two flow levels give four conditions, with five stream channels in each. NP = no pollution, P = pollution, LF = low flow, HF = high flow.

With two IVs of two levels each, this is a 2x2 design. Reading the schematic left to right also shows why the notation works the way it does. Crossing two levels with two levels *is* the multiplication that gives four conditions.

That four-cell grid resembles the chi-square table from Section 7.4.2, but the cells mean different things. There, the cells held *counts* of city blocks that already had a given land use and either did or did not top

40°C, and nobody put them there. Here, the cells are *conditions we assign channels to*, and within each one we then measure something else entirely, the species count.

Predictions. No pollution plus high flow should support the most species, and pollution plus low flow the fewest. The informative comparison is the gap between polluted and unpolluted channels at each flow rate. Those two gaps show whether pollution matters only when flow is low, with fast water washing its effect away, or hurts a stream no matter how fast the water moves. A one-factor study could not tell these apart, and the interaction test in a factorial experiment can.

Table 2 lists the species count in every channel.

Table 2: Macroinvertebrate species count in each of the 20 stream channels, five per condition.

Channel	No Pollution, Low Flow	Pollution, Low Flow	No Pollution, High Flow	Pollution, High Flow
1	10	5	12	9
2	8	4	13	8
3	11	3	14	10
4	9	4	11	11
5	10	2	13	12

Cell means and marginal means

Each condition, such as no pollution with low flow, is a cell, the same word Section 7.4.2 used for one combination in a table of counts. A **cell mean** is the mean of the scores in one cell. Here, each cell mean averages the species counts of the five stream channels assigned to that condition. A **marginal mean** is the mean for one level of a factor, averaged across the levels of the other factor. For example, the no-pollution marginal mean averages the two no-pollution cells, one at each flow rate. Table 3 shows the means from the experiment this chapter analyzes.

Table 3: Mean macroinvertebrate species count. The four inner numbers are cell means. The last row and column hold the marginal means, and the corner is the grand mean.

	Low Flow	High Flow	Marginal mean
No Pollution	9.6	12.6	11.1
Pollution	3.6	10.0	6.8
Marginal mean	6.6	11.3	8.95

The pollution marginal means (11.1 and 6.8) average each row across both flow rates. The flow marginal means (6.6 and 11.3) average each column across both pollution conditions. The corner, 8.95, is the grand mean of all 20 channels.

Why factorial designs matter

Suppose we ignored flow entirely and just compared polluted to unpolluted channels. That compares the two pollution marginal means, 11.1 vs. 6.8 species, shown in Figure 2.

A factorial design lets us ask whether the effect of pollution itself depends on flow. With flow rate added as a second IV, there are four cell means instead of two, so we can compare polluted and unpolluted channels *within* each flow rate, shown in Figure 3.

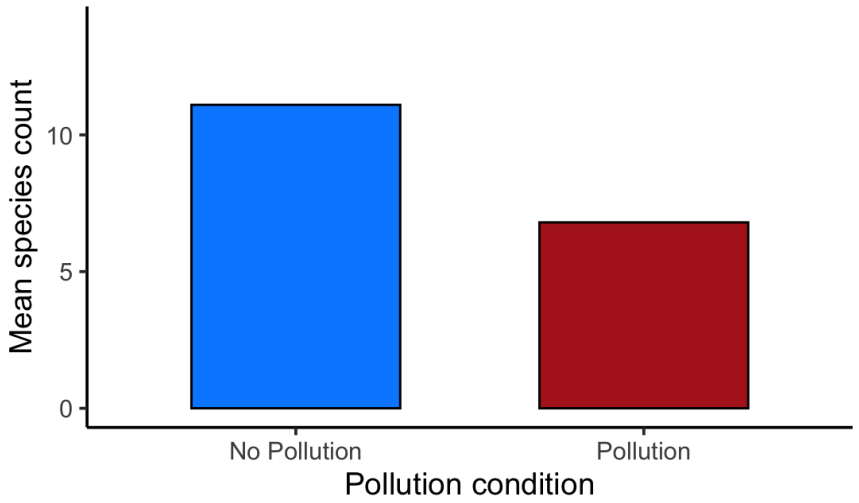


Figure 2: Mean macroinvertebrate species count under no pollution vs. pollution, averaged across flow rate (the pollution marginal means).

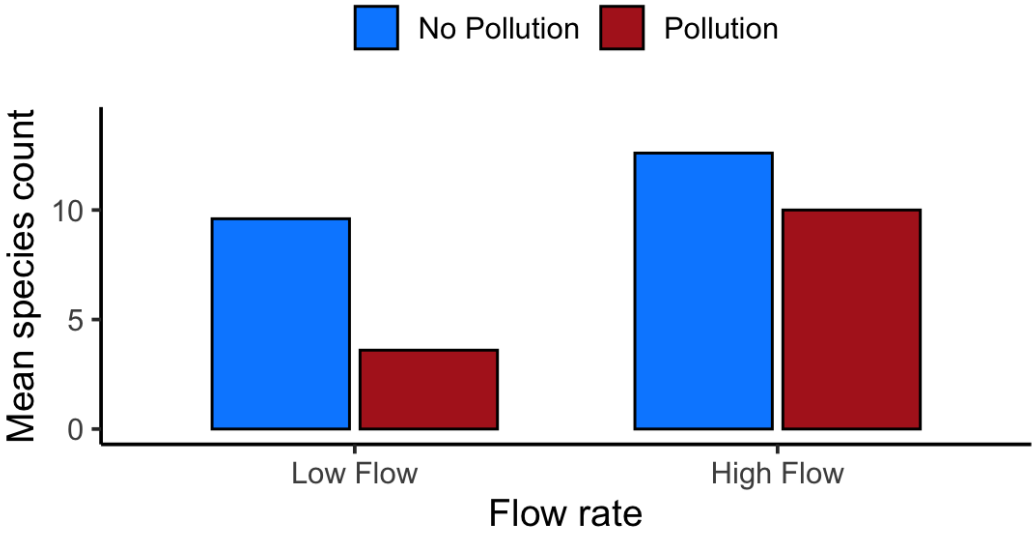


Figure 3: Mean macroinvertebrate species count in each of the four conditions, grouped by flow rate. The effect of pollution within each flow rate (the gap between the two bars) is larger under low flow than under high flow.

Under low flow, species count drops sharply with pollution (9.6 to 3.6, a difference of 6.0). Under high flow, the drop is smaller (12.6 to 10.0, a difference of 2.6). The effect of pollution depends on flow rate.

Main effects and interactions

A factorial design gives you two kinds of results: main effects and interactions. How many of each depends on the number of IVs.

A **main effect** is a difference among the marginal means of one IV. Each IV has one main effect, so a 2x2 design has two, one for Pollution and one for Flow. A 2x2x2 design and a 3x3x3 design both have three main effects, because the count depends on the number of IVs, not their levels.

A **simple effect** is the effect of one IV at a single level of the other IV, such as the pollution effect under low flow. Figure 3 shows two of them, one per flow rate, and the main effect of pollution is their average: $(6.0 + 2.6)/2 = 4.3$, the same 4.3 you get by subtracting the pollution marginal means.

An **interaction** occurs when the effect of one IV depends on the level of another. Figure 3 shows one in the sample means, since the simple effect of pollution changed with flow. The interaction test later in this chapter says whether the change is larger than chance would produce.

Equivalently, an interaction is a difference between differences. Each simple effect of pollution is a difference (unpolluted minus polluted). If that difference itself changes across flow rates, that's an interaction. The interaction sum of squares later in this chapter measures this difference between differences.

Table 4 collects the four quantities a 2x2 design gives, with the stream experiment's value for each.

Table 4: Cell mean, main effect, simple effect and interaction in the stream experiment.

Term	Definition	Stream experiment
Cell mean	The mean of the channels in one condition	No pollution with low flow: 9.6 species
Main effect	The difference between one factor's marginal means, averaged across the levels of the other factor	Pollution: $11.1 - 6.8 = 4.3$ species
Simple effect	The effect of one factor at a single level of the other	Pollution under low flow: 6.0 species. Under high flow: 2.6 species.
Interaction	The effect of one factor depends on the level of the other. The simple effects differ, a difference between differences.	$6.0 - 2.6 = 3.4$ species

The number of possible interactions scales with the number of IVs. A 2x2 design has exactly one interaction, Pollution × Flow. A design with three IVs (A, B, C) has three interactions between two of the IVs (A×B, A×C, B×C) and one interaction among all three (A×B×C).

What a 2x2 tests

A factorial ANOVA with two factors is also called a **two-way ANOVA**. It tests three effects, and each has its own null hypothesis:

1. a main effect of pollution,
2. a main effect of flow, and
3. a pollution × flow interaction.

The hypotheses are about the population mean species count for each condition, written μ. The cell means computed from the 20 channels, written $\bar{Y}$ later in this chapter, estimate them. Table 5 writes the four conditions and their mean species counts in this notation.

Table 5: The four conditions and their population mean species counts.

Pollution	Flow	Mean species count
No Pollution	Low Flow	$\mu_{\text{NoPoll,Low}}$
No Pollution	High Flow	$\mu_{\text{NoPoll,High}}$
Pollution	Low Flow	$\mu_{\text{Poll,Low}}$
Pollution	High Flow	$\mu_{\text{Poll,High}}$

A marginal population mean averages the two cell means it covers, so $\mu_{\text{NoPoll}} = (\mu_{\text{NoPoll,Low}} + \mu_{\text{NoPoll,High}})/2$, and μ_{Poll}, μ_{Low} and μ_{High} are found the same way.

Hypotheses

Main effect of pollution

- **Null (H_0):** Averaged across flow rates, mean species count is the same with and without pollution ($\mu_{\text{NoPoll}} = \mu_{\text{Poll}}$).
- **Alternative (H_a):** Averaged across flow rates, mean species count differs with and without pollution ($\mu_{\text{NoPoll}} \neq \mu_{\text{Poll}}$).

Main effect of flow

- **Null (H_0):** Averaged across pollution conditions, mean species count is the same at low and high flow ($\mu_{\text{Low}} = \mu_{\text{High}}$).
- **Alternative (H_a):** Averaged across pollution conditions, mean species count differs between low and high flow ($\mu_{\text{Low}} \neq \mu_{\text{High}}$).

Pollution × flow interaction

- **Null (H_0):** The effect of pollution is the same at both flow rates, $(\mu_{\text{NoPoll,Low}} - \mu_{\text{Poll,Low}}) = (\mu_{\text{NoPoll,High}} - \mu_{\text{Poll,High}})$.
- **Alternative (H_a):** The effect of pollution differs across flow rates.

Under the interaction null, the gap between polluted and unpolluted channels is the same whether the water is moving fast or slow. If that gap changes, for example if pollution does far more damage under low flow, you've found an interaction.

> **Note:** The interaction null for any two factors
>
> You'll sometimes see this written in abstract terms, with "Factor A" and "Factor B." For example,
>
> $$H_0 : (\mu_{A1,B1} - \mu_{A2,B1}) = (\mu_{A1,B2} - \mu_{A2,B2})$$
>
> The logic is the same, with A and B standing for your own variables.

The interaction in the cell means

An interaction is a difference between differences. Table 3 gives the four cell means to compute it with, before any sums of squares. Writing a simple effect as Δ, with the level it is computed at after the bar:

$$\Delta_{\text{Pollution|Low}} = \bar{Y}_{\text{NoPoll,Low}} - \bar{Y}_{\text{Poll,Low}} = 9.6 - 3.6 = 6.0$$

$$\Delta_{\text{Pollution|High}} = \bar{Y}_{\text{NoPoll,High}} - \bar{Y}_{\text{Poll,High}} = 12.6 - 10.0 = 2.6$$

Those two numbers aren't the same, so the effect of pollution isn't constant across flow rates, which is exactly what an interaction means. The gap between them, $6.0 - 2.6 = 3.4$, is a first, informal look at the interaction itself. The formal test below tells us whether a gap that size is larger than chance would produce.

Computing a factorial ANOVA by hand

In our between-subjects 2x2 design, 5 stream channels are assigned to each condition, 20 channels total, and no channel appears in more than one condition. The logic is the same as the one-way ANOVA in Chapter 8, which partitions the total variance into parts. For a 2x2 design, there's now one more part, the interaction:

$$SS_{\text{Total}} = SS_{\text{Pollution}} + SS_{\text{Flow}} + SS_{\text{Interaction}} + SS_{\text{Error}}$$

The tables below build up each piece. The first three have one row per channel, showing what that channel's score gets replaced with, its distance from the grand mean, and that distance squared.

SS Total

SS_{Total} captures all of the variation in the data. It squares the difference between each channel's score and the grand mean and adds them up. Table 6 shows the first few channels and the last, since the pattern is the same all the way down.

Table 6: Each channel's deviation from the grand mean, squared and summed.

Pollution	Flow	Species	Deviation from grand mean	Squared deviation
No Pollution	Low Flow	10	1.05	1.1025
No Pollution	Low Flow	8	-0.95	0.9025
No Pollution	Low Flow	11	2.05	4.2025
…	…	…	…	…
Pollution	High Flow	12	3.05	9.3025
Sums		179	0	242.95
Means		8.95		

The mean of all 20 scores is the grand mean, 8.95. The deviation column holds every channel's distance from it, and the last column squares those distances so they don't cancel (the same reason we squared deviations for the one-way ANOVA). Summing the squared column gives $SS_{\text{Total}} = 242.95$.

SS Pollution

$SS_{\text{Pollution}}$ isolates the variation we can explain by pollution condition alone. To compute it, replace every score with its pollution marginal mean, then find that mean's distance from the grand mean, as in Table 7.

Table 7: Each channel's score replaced by its pollution marginal mean, with that mean's deviation from the grand mean, squared and summed.

Pollution	Flow	Species	Pollution marginal mean	Marginal mean minus grand mean	Squared
No Pollution	Low Flow	10	11.1	2.15	4.6225
No Pollution	Low Flow	8	11.1	2.15	4.6225
No Pollution	Low Flow	11	11.1	2.15	4.6225
…	…	…	…	…	…
Pollution	High Flow	12	6.8	-2.15	4.6225
Sums		179		0	92.45

Every no-pollution channel is replaced by the no-pollution marginal mean, 11.1, and every pollution channel by the pollution marginal mean, 6.8. After that replacement, the only differences left among the 20 scores are the differences between the two pollution conditions, so their spread around the grand mean is the variation pollution explains. Every distance is either 2.15 or -2.15, so every squared distance is the same, and $SS_{\text{Pollution}} = 20 \times 2.15^2 = 92.45$.

SS Flow

SS_{Flow} works the same way, grouping by flow rate instead of pollution condition (Table 8).

Table 8: Each channel's score replaced by its flow marginal mean, with that mean's deviation from the grand mean, squared and summed.

Pollution	Flow	Species	Flow marginal mean	Marginal mean minus grand mean	Squared
No Pollution	Low Flow	10	6.6	-2.35	5.5225
No Pollution	Low Flow	8	6.6	-2.35	5.5225
No Pollution	Low Flow	11	6.6	-2.35	5.5225
…	…	…	…	…	…
Pollution	High Flow	12	11.3	2.35	5.5225
Sums		179		0	110.45

Every low-flow channel is replaced by the low-flow marginal mean, 6.6, and every high-flow channel by the high-flow marginal mean, 11.3. Every distance from the grand mean is either -2.35 or 2.35, so $SS_{\text{Flow}} = 20 \times 2.35^2 = 110.45$.

SS Interaction

The interaction is the one new piece a factorial ANOVA adds. It asks whether the four cell means differ from what the two main effects alone would predict.

If the main effects told the whole story, each cell mean would be the grand mean, shifted by that cell's pollution effect and its flow effect. Take the no-pollution, low-flow cell.

$$\begin{aligned}\text{No-pollution shift} &= 11.1 - 8.95 = +2.15\\ \text{Low-flow shift} &= 6.6 - 8.95 = -2.35\\ \text{Predicted cell mean} &= 8.95 + 2.15 - 2.35 = 8.75\\ \text{Observed cell mean} &= 9.6\\ \text{Leftover} &= 9.6 - 8.75 = 0.85\end{aligned}$$

The leftover is the part of that cell the two main effects can't explain. Those leftovers, one per cell, are what the interaction captures.

Each cell's predicted mean is the grand mean plus the pollution shift plus the flow shift, which simplifies:

$$\begin{aligned}\text{Predicted} &= \bar{Y}_{\text{grand}} + (\bar{Y}_{\text{Pollution marginal}} - \bar{Y}_{\text{grand}}) + (\bar{Y}_{\text{Flow marginal}} - \bar{Y}_{\text{grand}})\\ &= \bar{Y}_{\text{Pollution marginal}} + \bar{Y}_{\text{Flow marginal}} - \bar{Y}_{\text{grand}}\end{aligned}$$

Subtracting that from the cell mean gives each cell's leftover, which every channel in that cell shares:

$$\bar{Y}_{\text{cell}} - \bar{Y}_{\text{Pollution marginal}} - \bar{Y}_{\text{Flow marginal}} + \bar{Y}_{\text{grand}}$$

Table 9: Each cell's interaction leftover, found as the cell mean minus its pollution and flow marginal means plus the grand mean, then squared and counted once for each of the cell's five channels.

Cell	Cell mean	Minus pollution marginal mean	Minus flow marginal mean	Plus grand mean	Left-over	Leftover squared	Times 5 channels
No Pollution, Low Flow	9.6	11.1	6.6	8.95	+0.85	0.7225	3.6125
No Pollution, High Flow	12.6	11.1	11.3	8.95	-0.85	0.7225	3.6125
Pollution, Low Flow	3.6	6.8	6.6	8.95	-0.85	0.7225	3.6125
Pollution, High Flow	10.0	6.8	11.3	8.95	+0.85	0.7225	3.6125
Sum							14.45

In a balanced 2x2, the four leftovers are the same size, two positive and two negative (Table 9), so every channel's squared leftover is 0.85^2. Summing over the 20 channels gives $SS_{\text{Interaction}} = 20 \times 0.85^2 = 14.45$.

Each leftover is a quarter of the difference between differences computed earlier, $3.4/4 = 0.85$, so a larger gap between the two simple effects gives larger leftovers and a larger $SS_{\text{Interaction}}$.

SS Error

The last piece follows by subtraction, since every other term in the identity is already known:

$$SS_{\text{Error}} = SS_{\text{Total}} - SS_{\text{Pollution}} - SS_{\text{Flow}} - SS_{\text{Interaction}}$$

$$SS_{\text{Error}} = 242.95 - 92.45 - 110.45 - 14.45 = 25.6$$

SS_{Error} is the same quantity as SS_{Within} in the one-way ANOVA (Section 8.4.4), the spread of channels around their own cell mean. We can check it directly. The squared distances of the channels from their own cell mean add up within each cell:

$$\begin{aligned}
\text{No pollution, low flow} &= 0.4^2 + 1.6^2 + 1.4^2 + 0.6^2 + 0.4^2 = 5.2 \\
\text{Pollution, low flow} &= 1.4^2 + 0.4^2 + 0.6^2 + 0.4^2 + 1.6^2 = 5.2 \\
\text{No pollution, high flow} &= 0.6^2 + 0.4^2 + 1.4^2 + 1.6^2 + 0.4^2 = 5.2 \\
\text{Pollution, high flow} &= 1^2 + 2^2 + 0^2 + 1^2 + 2^2 = 10.0 \\
SS_{\text{Error}} &= 5.2 + 5.2 + 5.2 + 10.0 = 25.6
\end{aligned}$$

It is the variation none of the three effects can explain.

Checking the assumptions

A factorial ANOVA has the same test-specific assumptions as the one-way ANOVA (Section 8.5.2): homogeneity of variance (similar spread in every cell) and approximately normal residuals. Here a residual is a channel's count minus its own cell mean, the same deviations that make up SS_{Error}. Check them the same way, before reading any F, with Levene's test across the four cells and a Q-Q plot or Shapiro-Wilk test on the residuals. The residuals come from the fitted model, so in R we enter the data and fit the model first, then set its table aside until the checks are done.

```
all_df <- data.frame(
  Number_species = c(10, 8, 11, 9, 10,  5, 4, 3, 4, 2,
                     12, 13, 14, 11, 13,  9, 8, 10, 11, 12),
  Pollution = rep(c("No Pollution", "Pollution", "No Pollution", "Pollution"), each = 5),
  Flow      = rep(c("Low Flow", "Low Flow", "High Flow", "High Flow"), each = 5)
)
aov_fit <- aov(Number_species ~ Pollution * Flow, data = all_df)
car::leveneTest(Number_species ~ Pollution * Flow, data = all_df)
shapiro.test(residuals(aov_fit))
```

For the stream data, Levene's test across the four cells gives $F(3, 16) = 0.29$, $p = 0.835$, and the Shapiro-Wilk test on the residuals gives $W = 0.94$, $p = 0.263$, so neither check gives evidence of a problem.

Degrees of freedom, mean squares, and *F*

Each sum of squares becomes a mean square the same way it did in the one-way ANOVA (Section 8.4.6), by dividing by its degrees of freedom. With a levels of pollution, b levels of flow, and N channels in all:

- $df_{\text{Pollution}} = a - 1 = 1$
- $df_{\text{Flow}} = b - 1 = 1$
- $df_{\text{Interaction}} = (a-1)(b-1) = 1$, the product of the two main effects' degrees of freedom
- $df_{\text{Error}} = N - ab = 20 - 4 = 16$, one lost for each of the four cell means
- $df_{\text{Total}} = N - 1 = 19$, and $1 + 1 + 1 + 16 = 19$

Each effect gets its own F, and all three share the same denominator:

$$F_{\text{Pollution}} = \frac{MS_{\text{Pollution}}}{MS_{\text{Error}}}$$

$$F_{\text{Flow}} = \frac{MS_{\text{Flow}}}{MS_{\text{Error}}}$$

$$F_{\text{Interaction}} = \frac{MS_{\text{Interaction}}}{MS_{\text{Error}}}$$

Here $MS_{\text{Error}} = 25.60/16 = 1.60$. Every effect has 1 degree of freedom, so each effect's MS equals its SS, and Table 10 gives each F.

Table 10: Each effect's F, computed by hand as its MS divided by MS_{Error}.

Effect	SS	df	MS	F
Pollution	92.45	1	92.45	57.78
Flow	110.45	1	110.45	69.03
Interaction	14.45	1	14.45	9.03
Error	25.60	16	1.60	

Each F is compared with the critical value for 1 and 16 degrees of freedom, $F_{crit}(1, 16) = 4.49$ at $\alpha = 0.05$, and all three are larger.

Check your work

R's `aov()` builds the same table directly. The `*` in the formula tells R to include both main effects and their interaction. R writes the interaction row as `Pollution:Flow` and the error row as `Residuals`.

```
summary(aov_fit)
```

With `all_df` and `aov_fit` from the assumption checks, this line reproduces the table below.

Source	df	SS	MS	F	p
Pollution	1	92.45	92.45	57.78	< 0.001
Flow	1	110.45	110.45	69.03	< 0.001

Source	df	SS	MS	F	p
Pollution:Flow	1	14.45	14.45	9.03	0.008
Residuals	16	25.60	1.60		

The SS column matches the hand-calculated values above: 92.45 for Pollution, 110.45 for Flow, 14.45 for the interaction, 25.6 for Error. All three effects are significant here, including the interaction we previewed informally. The difference in the pollution effect between the two flow rates is larger than chance would produce.

This example is balanced (Section 8.5.2), with five stream channels in every condition. With unequal numbers, two things change. First, R computes the sums of squares one term at a time in the order the terms appear in `aov()`, so changing the order can change the results. Second, the marginal means become estimates from the fitted model, called estimated marginal means, instead of simple averages of the scores. Balanced designs avoid both complications, so plan equal sample sizes when you can.

Effect size in a factorial design

The one-way ANOVA had one effect and so needed one effect size, $\eta^2 = SS_{\text{Between}}/SS_{\text{Total}}$ (Section 8.5.3). A 2x2 runs three tests, so it needs three effect sizes, and the sums of squares we just computed by hand are all the ingredients required.

The direct extension keeps SS_{Total} in the denominator for every effect:

$$\eta^2 = \frac{SS_{\text{Effect}}}{SS_{\text{Total}}}$$

Each effect then reports its share of *all* the variation in the study, so the three values plus error's share add to 1. Adding a second factor that changes the outcome adds that factor's differences between conditions to the spread of all the scores, so SS_{Total} grows. The larger denominator shrinks the η^2 of a factor whose own SS did not change, only because a second factor was studied alongside it.

Partial eta-squared (η_p^2) is the proportion of variation one effect explains, out of that effect's variation plus the error variation. Its denominator is that effect's SS plus SS_{Error}:

$$\eta_p^2 = \frac{SS_{\text{Effect}}}{SS_{\text{Effect}} + SS_{\text{Error}}}$$

Because each effect gets its own denominator, partial values do not sum to 1. They are always at least as large as the corresponding η^2, because $SS_{\text{Effect}} + SS_{\text{Error}}$ is never larger than SS_{Total}. Figure 4 draws both kinds of denominator for the stream data.

The distinction never came up in Chapter 8, because a one-way ANOVA has only one effect, so there are no other factors to set aside, and $SS_{\text{Effect}} + SS_{\text{Error}}$ *is* SS_{Total}. The two formulas give the same number there.

Table 11: Eta-squared and partial eta-squared for each effect in the 2x2 design.

Effect	SS	Eta-squared	Partial eta-squared
Pollution	92.45	0.381	0.783
Flow	110.45	0.455	0.812
Pollution × Flow	14.45	0.059	0.361

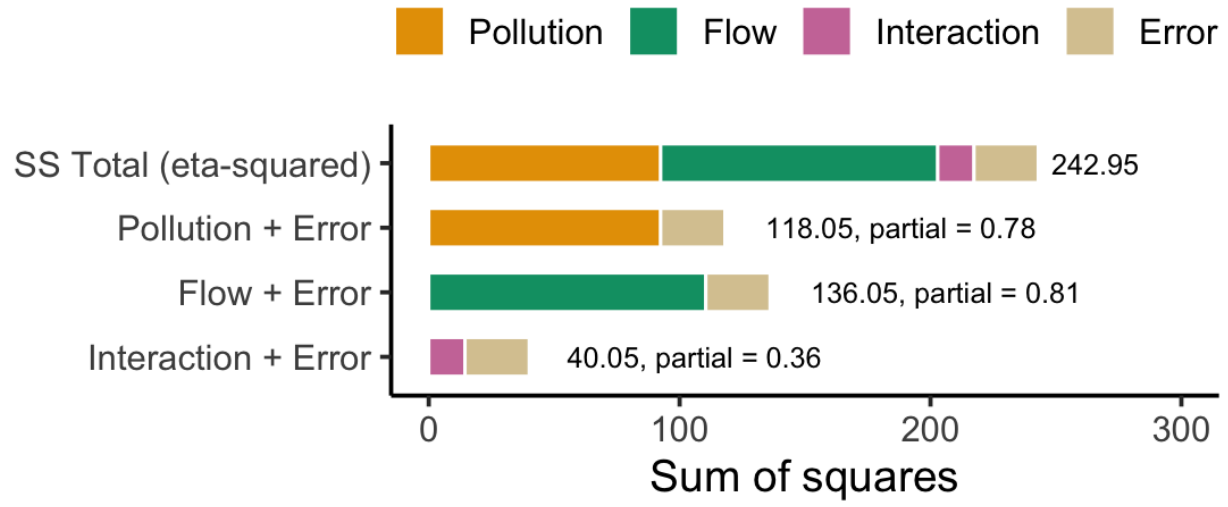


Figure 4: The denominators of eta-squared and partial eta-squared in the stream experiment, as sums of squares. The top bar is SS_{Total}, the denominator of every η^2. Each lower bar is one effect's SS plus SS_{Error}, the denominator of that effect's η_p^2. The label beside each lower bar gives its length and the effect's η_p^2.

Flow explains 45% of the total variation, pollution 38%, and the interaction 6%. The interaction is significant, and it is also the smallest of the three effects. Significance and size answer different questions (Chapter 5). A significant F says the effect is probably not zero, and η^2 says how large it is. Even with the smallest η^2, the interaction decides how the pollution main effect is read.

Say which one you report. Most statistical software, including SPSS and R's `effectsize` package, gives you η_p^2 by default, and it is the more common choice in published work. Either is defensible, but say which you used, because the two can differ substantially: pollution is $\eta^2 = 0.38$ but $\eta_p^2 = 0.78$ here. Reporting the larger number without labeling it overstates the effect. Do not compare η_p^2 from one study with η^2 from another, because the two use different denominators.

Interpreting main effects and interactions

A main effect is the difference between the marginal means of one IV, which average across the levels of the other factor. An interaction means the effect of one IV depends on the level of another. So when there is an interaction, each factor's main effect is an average of simple effects that differ across the levels of the other factor. In the stream data, pollution's simple effects (6.0 and 2.6) and flow's simple effects (3.0 and 6.4) both differ by 3.4 species.

Reading patterns in a 2x2

In any 2x2 ANOVA, each of the three effects is either present or absent, so there are $2 \times 2 \times 2 = 8$ possible outcome patterns, listed in Table 12. You don't need to memorize all eight, but you should learn to read them in a line plot.

Table 12: The eight possible patterns of effects in a 2x2. An X marks an effect that is present.

Pattern	Main Effect: Pollution	Main Effect: Flow	Interaction
1			
2	X		
3	X		X
4	X	X	
5	X	X	X
6		X	
7		X	X
8			X

Figure 5 shows the eight patterns as bar graphs, which show differences in means directly, and Figure 6 below shows the same patterns as line graphs, which make interactions easier to spot. These are abstract illustrations of every combination that can occur, so a few of them describe patterns you would not expect ecologically. Use them to learn to recognize the shapes.

From here on, every figure puts flow rate on the x-axis and draws one line (or bar) per pollution condition, blue for no pollution and red for pollution. In the line graphs, no pollution is solid and pollution is dashed.

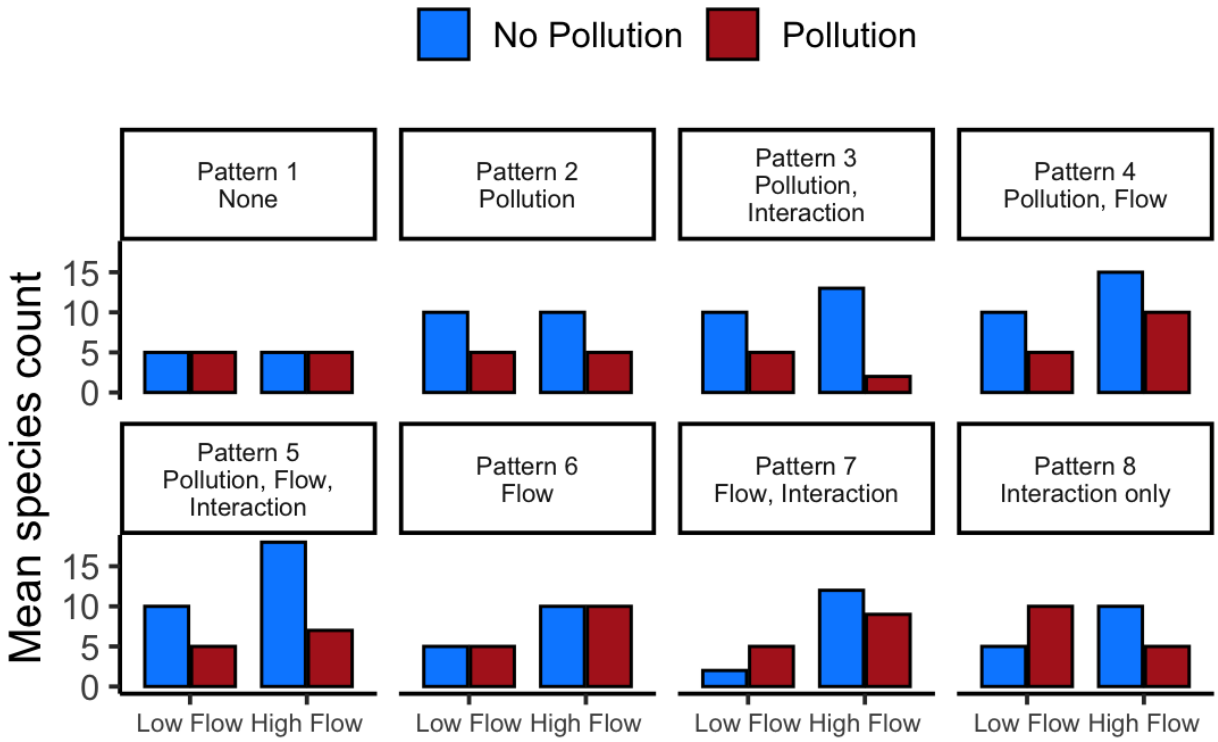


Figure 5: Eight example patterns of main effects and interactions in a pollution × flow design, numbered to match the rows of Table 12.

Figure 6 shows the same patterns in line graph form:

In the line graphs, interactions are easiest to see:

- **Parallel lines** mean no interaction (the pollution effect is the same at both flow rates).

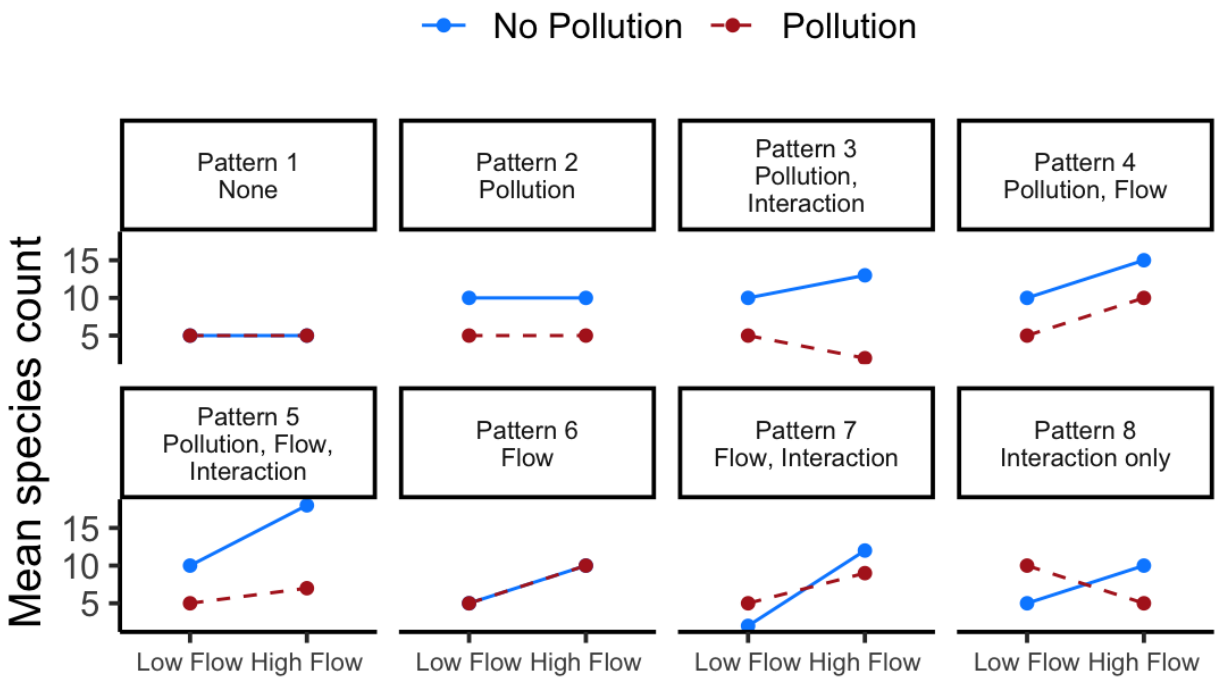


Figure 6: Line graphs showing eight possible outcomes for a pollution × flow design. In Patterns 1 and 6 the two lines lie on top of each other.

- **Non-parallel lines** mean an interaction (the pollution effect differs by flow). The lines may cross inside the plot, or they may converge or diverge, which means they would cross if you extended them.
- The average gap between the two lines is the main effect of pollution, and the lines' average rise or fall from low to high flow is the main effect of flow.

> Tip: Which factor goes on the x-axis?
>
> The statistics are identical either way, since the same cell means get plotted regardless. Two rules of thumb make the picture easier to read. Put the factor with more levels on the x-axis, since one long line is easier to follow than several short, crowded ones. And draw the factor whose effect you're asking about as the lines, so its simple effect at each level of the other factor is the gap between the lines. This chapter puts flow on the x-axis because the question is how the effect of pollution changes with flow.

Simple effects

When an interaction is present, the natural follow-up is to describe each simple effect, the effect of one factor at a single level of the other. Start with the Δ values from the cell means, as we did earlier:

$$\Delta_{\text{Pollution|Low}} = 9.6 - 3.6 = 6.0$$
$$\Delta_{\text{Pollution|High}} = 12.6 - 10.0 = 2.6$$

Their difference, 3.4 species, is the interaction. Pollution costs 6.0 species under low flow and 2.6 under high flow. But saying "pollution reduced species count under low flow" is still a claim, and like any claim in this book, it needs its own test, not just a look at the plot.

Testing a simple effect asks whether pollution's effect holds at one specific level of flow, so it looks like a two-sample comparison: no pollution vs. pollution, at low flow only. One option is to pull out the 10 low-flow channels and run a two-sample t-test on them alone. That test estimates the error variance from those 10 channels, with 8 degrees of freedom, and ignores the 10 high-flow channels. The factorial ANOVA computed earlier in this chapter already estimates that error variance from all 20 channels, as MS_{Error} with 16 degrees of freedom instead of 8.

At each flow rate the test has its own pair of hypotheses. At low flow:

- **Null (H_0):** At low flow, mean species count is the same with and without pollution, $\mu_{\text{NoPoll,Low}} = \mu_{\text{Poll,Low}}$.
- **Alternative (H_a):** At low flow, mean species count differs with and without pollution, $\mu_{\text{NoPoll,Low}} \neq \mu_{\text{Poll,Low}}$.

The simple-effects test reuses that pooled estimate, MS_{Error} with its 16 degrees of freedom, in a two-sample t:

$$t = \frac{\bar{Y}_{\text{NoPoll},j} - \bar{Y}_{\text{Poll},j}}{\sqrt{MS_{\text{Error}}\left(\frac{1}{n_{\text{NoPoll},j}} + \frac{1}{n_{\text{Poll},j}}\right)}}, \qquad df = df_{\text{Error}}$$

Here j is one flow rate, so the two means are the no-pollution and pollution cell means at that flow rate. Using $MS_{\text{Error}} = 1.60$ and $df = 16$ from the omnibus table computed earlier, the denominator is the same at both flow rates.

$$\sqrt{MS_{\text{Error}}\left(\tfrac{1}{5}+\tfrac{1}{5}\right)} = \sqrt{1.60(0.4)} = \sqrt{0.64} = 0.80$$

$$t_{\text{Low}} = \frac{6.0}{0.80} = 7.50$$

$$t_{\text{High}} = \frac{2.6}{0.80} = 3.25$$

Table 13 gives both tests.

Table 13: Simple effects of pollution, tested separately within each flow rate, using the pooled error term from the full model.

Simple effect	Difference	t	df	p
Pollution, within Low Flow	6.0	7.50	16	< 0.001
Pollution, within High Flow	2.6	3.25	16	0.005

Table 14 compares the two ways of testing the low-flow simple effect. Here the separate test gives the larger t, because the pollution, high-flow cell has the most spread (variance 2.5, against 1.3 in each of the other three), and that cell raises MS_{Error}. The pooled test still has twice the degrees of freedom, so its critical value is lower, and its error estimate rests on all 20 channels instead of 10.

Table 14: The low-flow simple effect of pollution, tested two ways. The separate test uses only the 10 low-flow channels. The pooled test uses the error mean square from all 20 channels.

Approach	Error variance	df	SE	t	Critical t
Separate t-test, low-flow channels only	1.30	8	0.72	8.32	2.31
Pooled, error mean square from the full model	1.60	16	0.80	7.50	2.12

Both comparisons in Table 13 clear the critical value, $t_{crit}(16) = 2.12$ at $\alpha = 0.05$, two-tailed, with $t = 7.50$ under low flow and $t = 3.25$ under high flow. Both rows are significant, so pollution lowered species count at both flow rates, and the interaction says the drop was larger under low flow (6.0 species) than under high flow (2.6 species). If the high-flow row had not been significant, that would mean the test could not detect a pollution effect at high flow. It would not show that the effect there is zero (Section 8.5.4).

When a factor has only two levels, as Pollution does here, each simple-effects test compares those two levels, so it is the full answer. When a factor has three or more levels, a significant simple effect says only that some levels differ at that level of the other factor, and pairwise comparisons are needed to say which.

Order of interpretation

When we conduct a two-way ANOVA, we always begin by reading the interaction term, because it decides what the main effects mean.

- If the interaction is *not significant*, examine the two main effects separately, interpreting each as an overall difference averaged across the other factor. If a significant main effect has three or more levels, follow it with Tukey HSD on that factor, exactly as in Section 8.5.4.

- If the interaction is significant, lead with the simple effects: how one factor behaves at each level of the other. "Pollution reduced species count sharply under low flow, and less under high flow" is a simple-effects sentence. "There was a main effect of pollution" on its own is not enough. Whether the main effect is still worth reporting depends on the direction of the simple effects:
  - If the simple effects all point the same way and differ only in size, the main effect gives the direction of the effect at every level of the other factor. Report it, qualified by the interaction: "There was a main effect of pollution, qualified by a Pollution × Flow interaction."
  - If one simple effect is near zero, or they point in opposite directions, the main effect is an average that describes no actual condition. Don't report it. Report the simple effects only.
- If *neither* the interaction nor either main effect is significant, there is nothing to follow up on. Do not run, or interpret, post-hoc tests on a result that never cleared the omnibus test in the first place.

The three examples below all have an interaction. They show when a main effect can still be reported and when it cannot.

A consistent main effect and an interaction

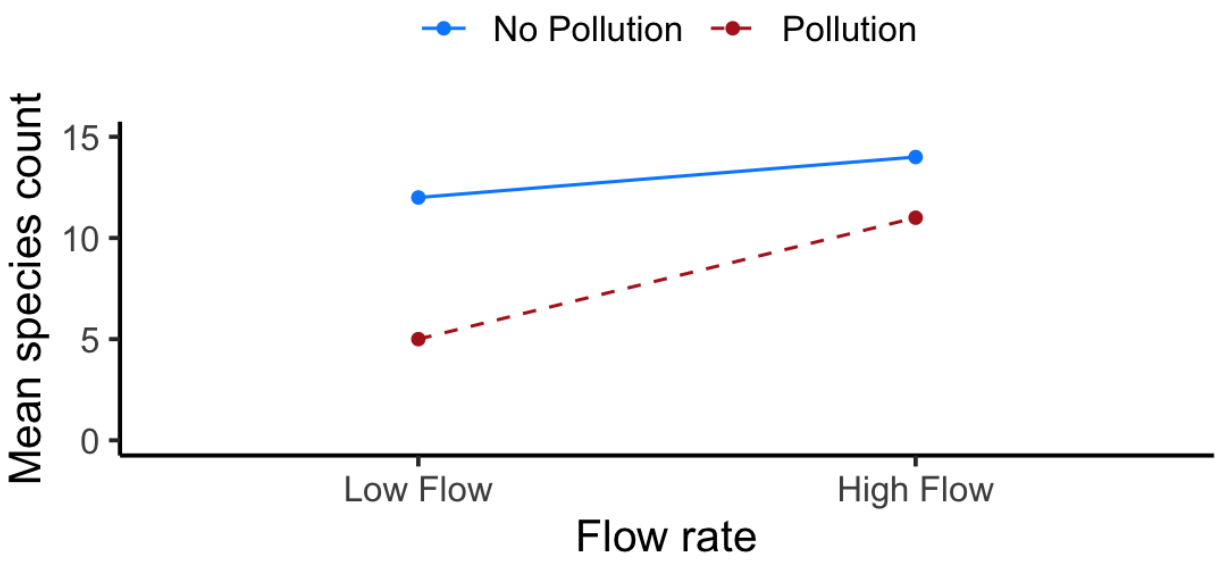


Figure 7: Pollution shows a main effect, but the size of that effect differs by flow rate (interaction).

Figure 7 shows both a main effect and an interaction. Unpolluted channels are above polluted ones at both flow rates. That is the main effect. But the size of that gap is larger under low flow (12 vs. 5) than under high flow (14 vs. 11), so there's an interaction too.

If an ANOVA on data with these cell means found both effects significant, we would report a main effect of pollution, qualified by a Pollution × Flow interaction. The size of the pollution effect changes with flow rate, larger under low flow and smaller under high flow. The main effect still describes both flow rates, because pollution is associated with lower species counts at each. Flow changes only how much lower.

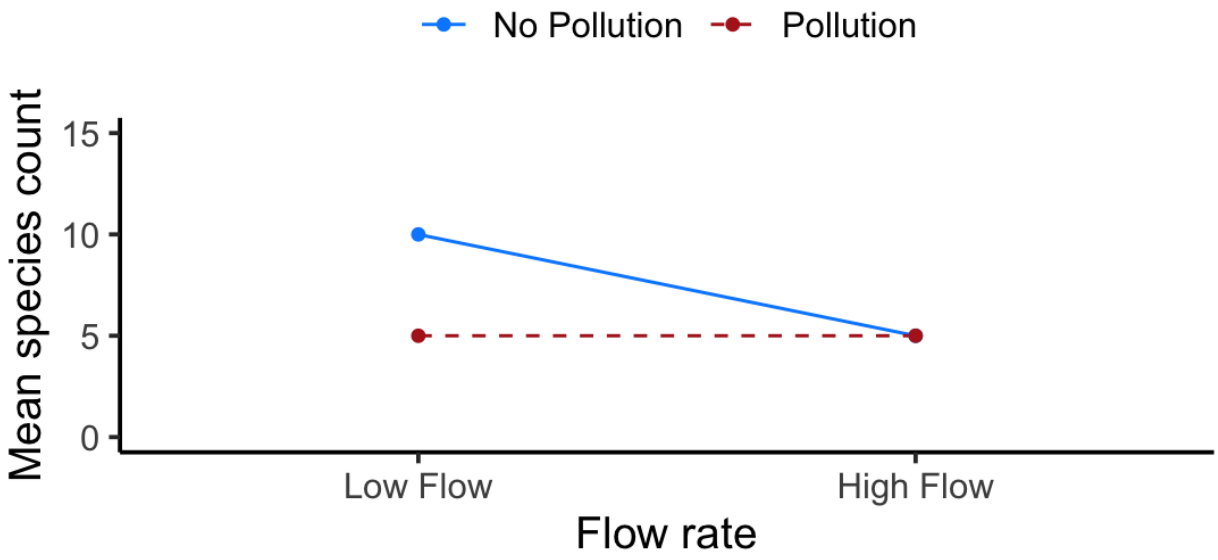


Figure 8: An apparent main effect of pollution arises from averaging over an interaction.

An "averaged" main effect driven by an interaction

Figure 8 shows a starker pattern, with a large difference between polluted and unpolluted channels under low flow and no difference at all under high flow.

Yet an ANOVA on data with these cell means can still report a significant main effect of pollution when the error variance is small. Averaged across flow rates, unpolluted channels give $(10 + 5)/2 = 7.5$ species and polluted channels give 5, a main-effect difference of 2.5. That number is correct arithmetic, but it matches the pollution effect at neither flow rate. Pollution removes 5 species under low flow and none under high flow, not 2.5 everywhere. The "main effect" here is an artifact of averaging over a strong interaction.

If the ANOVA did show a significant main effect of pollution, it should not be reported, because the pollution effect was large under low flow and zero under high flow. Report the simple effects instead. In a case like this, the cell means give no support for a pollution effect at every flow rate, because pollution lowered species count only under low flow.

A main effect masked by an interaction

Figure 9 shows the most misleading pattern of the three. Under low flow, pollution *reduces* species count (8 vs. 4). Under high flow, pollution *increases* it (8 vs. 12). Both effects are 4 species, in opposite directions.

Average across flow rates to get a main effect, and the two effects cancel:

$$\begin{aligned} \text{Unpolluted} &= (8 + 8)/2 = 8 \\ \text{Polluted} &= (4 + 12)/2 = 8 \end{aligned}$$

The main-effect difference is exactly 0. An ANOVA on data with these cell means would report no main effect of pollution at all, because its two opposite effects average each other out.

Reading the interaction first prevents this mistake. A non-significant main effect here would be true and misleading. Pollution has no effect *on average* and a large effect *in each condition*. Reporting only the

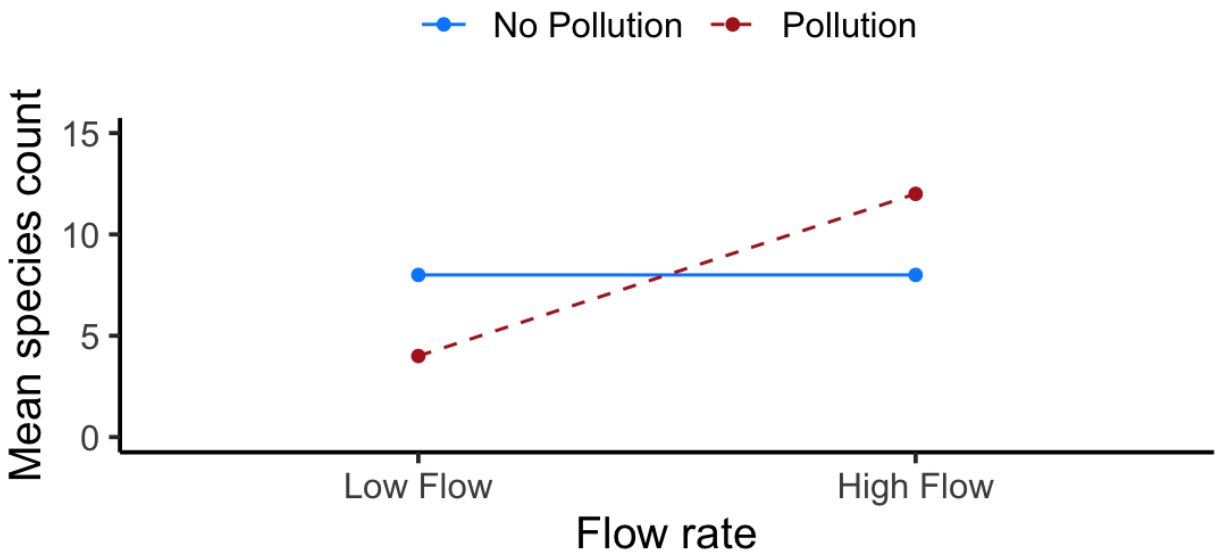


Figure 9: Pollution's effect reverses direction between flow rates, so the main effect averages to zero even though pollution matters at both.

main effect would suggest pollution is irrelevant, when it has a large effect at both flow rates, in opposite directions.

Table 15 puts the three cases and the stream experiment side by side.

Table 15: The same rule applied to four sets of cell means.

Case	Simple effect of pollution (low flow, high flow)	Main effect of pollution	What to report
Consistent main effect and an interaction (Figure 7)	7, 3	5	Simple effects, plus the main effect with its qualification
Averaged main effect (Figure 8)	5, 0	2.5, which matches neither condition	Simple effects only
Masked main effect (Figure 9)	4, −4	0, which hides two effects of 4 species	Simple effects only
Stream experiment (Figure 10)	6.0, 2.6	4.3	Simple effects, plus the main effect with its qualification

Reporting the stream result

Back to the stream experiment. Figure 10 plots its four cell means. Pollution lowers species count at both flow rates, so the lines do not cross inside the plot. They converge, because the gap under high flow is smaller. This matches the first case in Table 15, a consistent main effect and an interaction, so we report the simple effects and the main effect of pollution with its qualification attached.

The same check applies to flow. Flow raised species count in both pollution conditions, by $12.6 - 9.6 = 3.0$ species without pollution and $10.0 - 3.6 = 6.4$ with it, so the flow main effect can also be reported with its qualification.

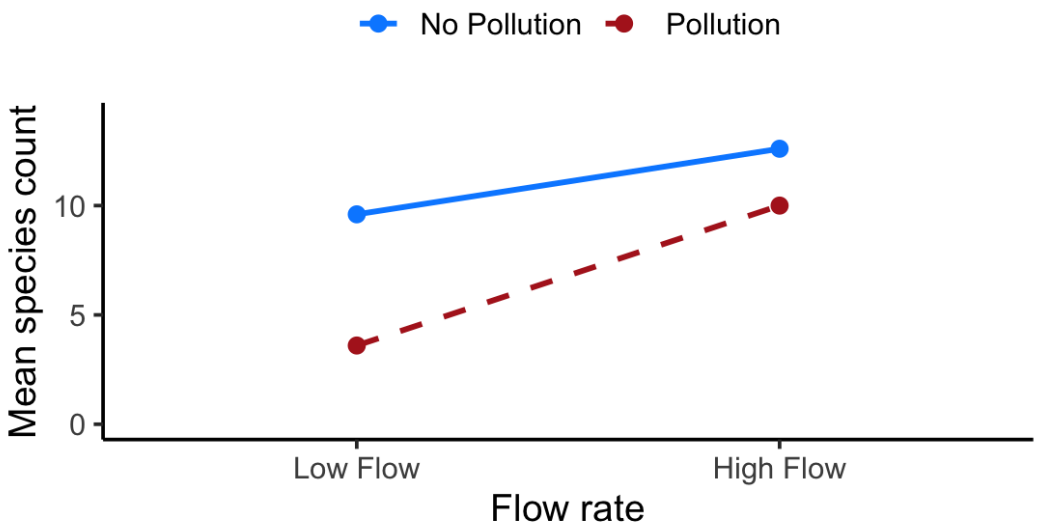


Figure 10: Cell means from the stream experiment. Pollution lowers species count at both flow rates, by more under low flow.

! Results reporting: factorial ANOVA

Lead with the interaction, then give the simple effects. Report a main effect only when the simple effects point the same way, and then with its qualification attached. Every effect gets its own effect size, labeled so the reader knows whether it is η^2 or η_p^2.

> The effect of pollution on macroinvertebrate species count depended on flow rate, $F(1, 16) = 9.03$, $p = 0.008$, $\eta_p^2 = 0.36$. Under low flow, polluted channels supported 6.0 fewer species than unpolluted ones (3.6 vs. 9.6), $t(16) = 7.50$, $p < 0.001$. Under high flow that gap narrowed to 2.6 species (10.0 vs. 12.6), $t(16) = 3.25$, $p = 0.005$. Both main effects were also significant, pollution $F(1, 16) = 57.78$, $p < 0.001$, $\eta_p^2 = 0.78$, and flow $F(1, 16) = 69.03$, $p < 0.001$, $\eta_p^2 = 0.81$, but each should be read in light of the interaction. Pollution cost a stream more species when the water moved slowly (6.0 vs. 2.6).

Table 16 summarizes the two-factor ANOVA.

Table 16: Summary of the two-factor ANOVA.

Purpose	Do two factors, crossed in every combination, each change the mean?
	Does the effect of one depend on the level of the other?
Hypotheses	No main effect of A, $H_0 : \mu_{A1} = \mu_{A2} = \cdots = \mu_{Aa}$ (averaged over B)
	No main effect of B, $H_0 : \mu_{B1} = \mu_{B2} = \cdots = \mu_{Bb}$ (averaged over A)
	No interaction (2x2 case), $H_0 : (\mu_{A1,B1} - \mu_{A2,B1}) = (\mu_{A1,B2} - \mu_{A2,B2})$
	Each H_a says its null is false
Assumptions	Similar spread in every cell
	Approximately normal residuals

Test statistic	$F = MS_{\text{Effect}}/MS_{\text{Error}}$ for each effect $df_A = a - 1$ $df_B = b - 1$ $df_{\text{Interaction}} = (a-1)(b-1)$ $df_{\text{Error}} = N - ab$
Report	Interaction first, with $F(df, df_{\text{Error}})$, p, and η_p^2 labeled as partial If it is significant, the simple effects, and a main effect only when the simple effects point the same way If it is not, each main effect, with Tukey HSD for a significant factor with three or more levels

More complicated designs

Up until now we have focused on the simplest case for factorial designs, the 2x2 design, with two IVs, each with 2 levels. The same logic extends to larger designs.

3x2 design

This design works like the 2x2, with one more level added to one factor. The test structure is the same (two main effects and one interaction).

The same stream example extends to a 3x2. Suppose we examine macroinvertebrate species count (the dependent variable, DV) under three flow rates (low, medium, high) crossed with two pollution conditions (no pollution vs. pollution).

- The main effect of pollution tests whether, on average, polluted and unpolluted channels differ in species count across all flow rates.
- The main effect of flow tests whether mean species count differs among flow rates.
- The interaction tests whether the pollution effect (unpolluted − polluted) changes with flow rate.

In a 2x2 design, you look for an interaction by checking whether the lines are parallel. In a 3x2, the idea is the same, but with three points per line the patterns are harder to read.

You have three hypotheses in a 3x2 factorial design:

Hypothesis 1: Main effect of pollution

- **Null (H_0):** Averaged across flow rates, mean species count is the same with and without pollution, $\mu_{\text{NoPoll}} = \mu_{\text{Poll}}$.
- **Alternative (H_a):** Mean species count differs between polluted and unpolluted channels.

Hypothesis 2: Main effect of flow

- **Null (H_0):** Averaged across pollution conditions, mean species count is the same at all three flow rates, $\mu_{\text{Low}} = \mu_{\text{Medium}} = \mu_{\text{High}}$.
- **Alternative (H_a):** At least one flow rate mean differs from the others.

Hypothesis 3: Pollution × flow interaction

- **Null (H_0):** The pollution effect is the same at all three flow rates, $(\mu_{\text{NoPoll,Low}} - \mu_{\text{Poll,Low}}) = (\mu_{\text{NoPoll,Medium}} - \mu_{\text{Poll,Medium}}) = (\mu_{\text{NoPoll,High}} - \mu_{\text{Poll,High}})$.
- **Alternative (H_a):** The pollution effect differs across flow rates, meaning at least one of the three pollution differences is not equal to the others.

One factorial ANOVA is better than a separate t-test of polluted vs. unpolluted channels at each of the three flow rates, for two reasons. First, the ANOVA estimates the error variance from all six cells, which gives more degrees of freedom and so more power for each test. Second, it gives a formal test of the interaction, which is often the scientific question.

Figure 11 shows example means for this design.

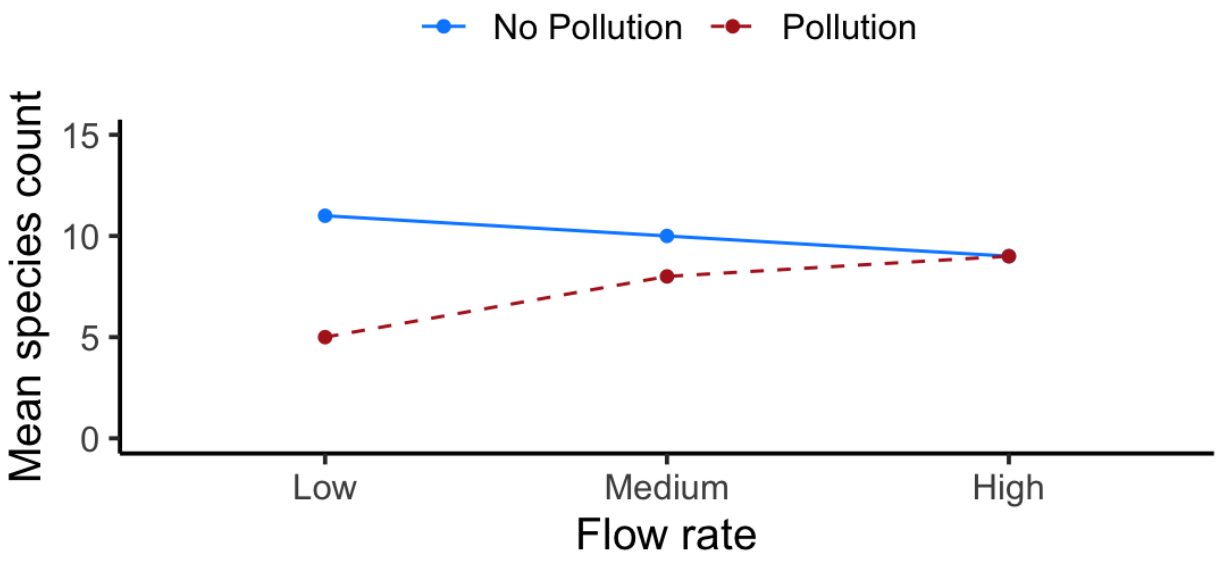


Figure 11: Example means for a 3x2 pollution × flow design.

Table 17 lists the six invented cell means, their marginal means, and the pollution gap at each flow rate.

Table 17: Mean species count in the 3x2 example. The last column and the marginal-mean row hold the marginal means, the corner is the grand mean, and the last row is the pollution gap (no pollution minus pollution) at each flow rate.

	Low	Medium	High	Marginal mean
No Pollution	11	10	9	10.0
Pollution	5	8	9	7.3
Marginal mean	8	9	9	8.67
Pollution gap	6	2	0	

1. **Main effect of pollution.** Averaged across flow rates, unpolluted channels support higher species counts (10 vs. 7.3). The “No Pollution” line is above the “Pollution” line at low and medium flow, and the two coincide at high flow.
2. **Main effect of flow.** Averaged over pollution condition, species counts are lowest at low flow (8) and higher at medium and high flow (9 each).

3. **Interaction between pollution and flow.** An interaction occurs when the effect of one IV depends on the levels of another. The pollution gap is 6 species at low flow, 2 at medium, and 0 at high. So the size of the pollution effect changes with flow rate, which is also why the pollution main effect above describes no single flow condition well. The interaction test in the ANOVA would say whether these differences are larger than chance would produce.

If there were no interaction and no main effect of flow, we would see something like the pattern in Figure 12.

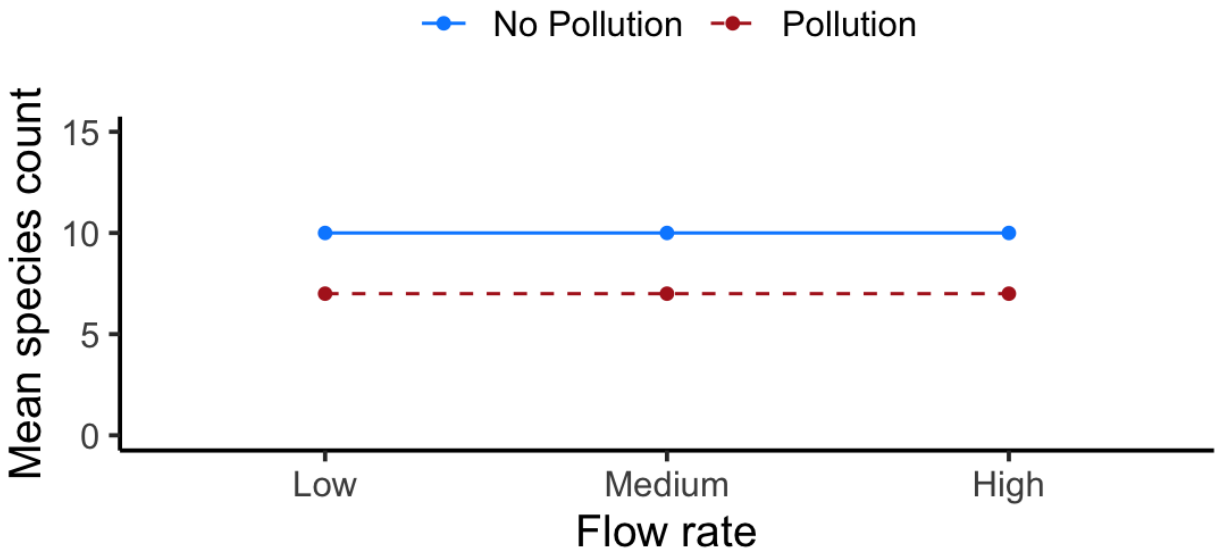


Figure 12: Example means for a 3x2 design with a main effect of pollution and no other effects.

Figure 13 shows a different interaction pattern.

The means show an interaction here too. The pollution effect is measured three times, once at each flow rate. It is the same at low and medium flow but much smaller at high flow, so its size depends on the level of flow.

Designs beyond this course

Larger factorial designs use the same logic as the 2x2. But complexity grows quickly, both in the number of effects and in how hard they are to interpret.

2x2x2 (three-factor) designs

A 2x2x2 design adds a third IV with two levels. Start from the two-way interaction in the stream experiment. Pollution lowered species count by 6.0 under low flow and by 2.6 under high flow, so the pollution × flow interaction is the difference between those two simple effects, $6.0 - 2.6 = 3.4$ species.

Now suppose the same experiment ran in summer and again in winter, so season is a third IV. Each season has its own 2x2 and its own pollution × flow interaction. A **three-way interaction** means that the two-way interaction changes across the third factor. If the interaction were 3.4 species in summer and close to 0 in winter, the pollution × flow interaction would depend on season, which is a three-way interaction. If both

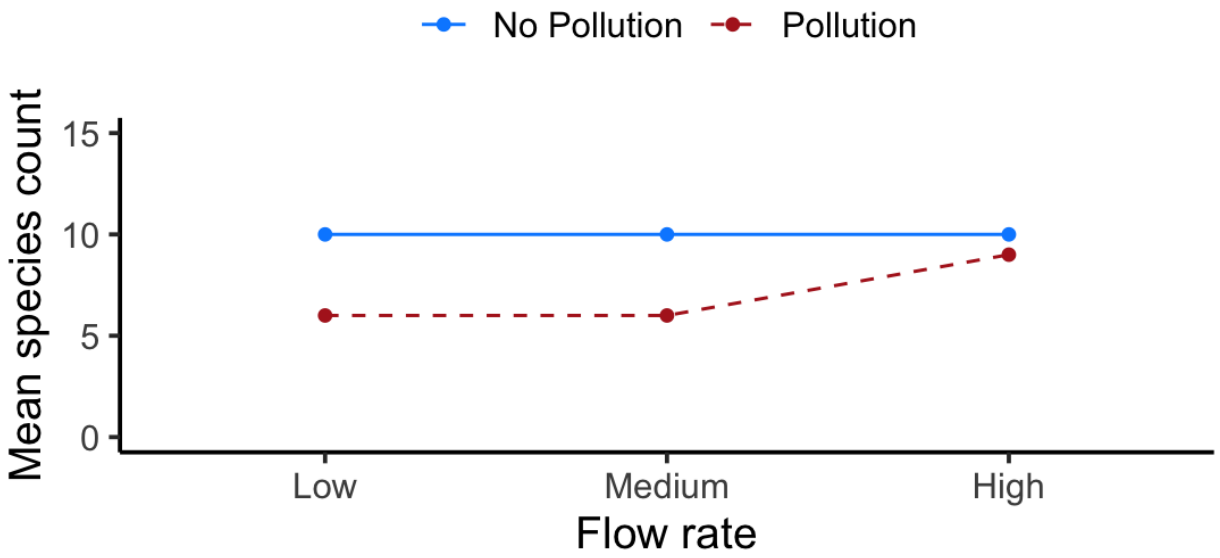


Figure 13: Example means for a 3x2 design in which the pollution gap is the same at low and medium flow and smaller at high flow.

seasons showed about 3.4, there would be no three-way interaction, even though each season has a two-way interaction. An interaction between two IVs is a **two-way interaction**, so a 2x2 has one and a design with three IVs has three. Table 18 lists all seven effects.

Table 18: The seven effects in a 2x2x2 design, using the stream experiment with season added.

Effect	How many	What it asks, with season added
Main effects	3	Does pollution change the mean? Flow? Season? Each is averaged across the other two IVs
Two-way interactions	3	Does the pollution effect depend on flow, or on season? Does the flow effect depend on season?
Three-way interaction	1	Does the pollution × flow interaction change between summer and winter?

Graph it as two 2x2 plots side by side, one for each season. A three-way interaction shows up as a different pattern of lines in the two plots. Each added factor multiplies the number of conditions and adds effects to interpret, so add a third factor only when the question needs it.

Mixed designs

A **mixed design** has at least one between-subjects factor and at least one within-subjects factor (Section 8.6). For example, sites or treatments could be a between-subjects factor, and repeated measures over time a within-subjects factor.

The interpretation of main effects and interactions is the same, but the error terms used in the F-tests differ, and the data structure must account for repeated measures. These models are important in longitudinal and environmental time-series studies, but they're beyond our current scope.

Chapter Summary

Why it matters. Single-factor designs can tell you whether something has an effect. Factorial designs tell you what *changes* that effect, which is usually the more interesting scientific question, and often the more realistic one for environmental systems where drivers act together. An interaction changes what the main effects mean, so the interaction row is read before the main-effect rows. Chapter 11 replaces one factor with a continuous covariate.

Core ideas

- **A fully factorial design measures every combination.** It pairs every level of each IV with every level of the others (fully crossed). The notation says the structure: 2x2 is four conditions, 3x2 is six.
- **A 2x2 tests three things.** One main effect per factor, plus their interaction. That means three null hypotheses, and you should be able to state all three in plain language before running anything.
- **An interaction is a difference between differences.** It asks whether the effect of one factor changes across levels of another. The interaction sum of squares measures it.
- **The partition extends the one-way logic.** SS_{Total} now splits into each main effect, the interaction, and error (SS_{Error}, Chapter 8's SS_{Within} in the one-way ANOVA). Each effect gets its own F, all sharing MS_{Error} as the denominator. Working one table through by hand shows that the interaction term is the only new piece.
- **Each test gets its own effect size.** η^2 divides each effect's SS by SS_{Total}, and η_p^2 divides it by that effect plus error. Partial values are at least as large and do not sum to 1, so label which one you are reporting. A significant effect can still be the smallest one, as this chapter's interaction is ($\eta^2 = 0.06$, against 0.38 for pollution and 0.45 for flow).
- **Read the interaction first.** If it's significant, lead with simple effects: what factor A does at each level of B, as Δ values from the cell means, tested with the pooled MS_{Error}. A main effect can still be reported, qualified, when the simple effects all point the same way. When one is near zero or they reverse, the main effect describes no actual condition. When the interaction is not significant, interpret each main effect as an overall difference, with Tukey HSD for a factor with three or more levels.
- **Learn to read the picture.** In a line plot, parallel lines mean no interaction, and lines that cross, inside the plot or if extended, mean an interaction. The average gap between the lines is one main effect, and the lines' average rise or fall is the other. There are eight possible patterns in a 2x2. The plot shows which pattern the cell means follow, and the interaction F test says whether the lines differ from parallel by more than chance would produce.
- **An averaged main effect can be an artifact.** A factor that matters a lot in one condition and not at all in another can still produce a "significant main effect" once you average across conditions. That number is correct arithmetic, but it matches the pollution effect at neither flow rate.
- **A 3x2 has the same three tests as a 2x2.** Its three points per line are harder to read than two. Beyond that, a three-factor design brings three main effects, three two-way interactions, and a three-way interaction, and interpretability falls off fast. Use only as many factors as your question actually needs.